

Intro to DAG

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Definition 0.0.1 (dga) *DG k -algebra is a graded vector space $A = \oplus A_i$ s.t.*

- $A_i \times A_j \rightarrow A_{i+j}$
- $1 \in A_0$
- $\delta : A_i \rightarrow A_{i-1}$ for all i , $\delta^2 = 0$, $\delta(a \cdot b) = \delta(a) \cdot b + (-1)^{\deg(a)} a \cdot \delta(b)$
- $a \cdot b = (-1)^{\deg(a)\deg(b)} b \cdot a$ (commutativity)

The morphism of cdga is $f : A \rightarrow B$ where $f \cdot \delta_A = \delta_B \cdot f$ and $f(a \cdot_A b) = f(a) \cdot_B f(b)$.

$dgAlg_k$ is category of cdga. The opposite category $(dgAlg_k)^{op}$ is category of affine dg-schemes, whose element we define as $Spec(A)$.

Definition 0.0.2 (Dold-Kan Correspondence) *Normalized chain complex functor $N : sAb \rightarrow Ch_{\geq 0}(Ab)$ is an equivalence of functor (different from nerve), and its simplicialization functor $\Gamma : Ch_{\geq 0}(Ab) \rightarrow sAb$ is as well. N is given by $N(A)_n = \bigcap_{i=1}^{\infty} \ker(d_i : A_n \rightarrow A_{n-1})$ with differential $d = d_0 : N(A)_n \rightarrow N(A)_{n-1}$.*

This defines the homotopy of simplicial commutative ring $\pi_i(A) = H_i(N(A))$.

Definition 0.0.3 (Simplicial Commutative Ring) *Simplicial commutative ring is a simplicial object $A : \Delta^{op} \rightarrow CRing$ in the category $CRing$, which means family of commutative rings A_i with face map $d_i : A_n \rightarrow A_{n-1}$ and degeneracy map $s_i : A_n \rightarrow A_{n+1}$.*

Definition 0.0.4 (E_{∞} -operad) *E_{∞} -operad is a topological operad where $E_{\infty}(k) \cong *$ is contractible for all k . Or that can be direct limit of E_n operad as $E_{\infty}(k) = \text{colim}_n E_n(k)$ as a topological space. The limit is clearly $E_1 \hookrightarrow E_2 \hookrightarrow E_3 \hookrightarrow \dots$.*

Definition 0.0.5 (dg-scheme) *(definition 1 : homology) spectrum is $Spec(H^0(A))$ where A is cdga, while $H^0(A) = \ker(A_1 \rightarrow A_0)$ is commutative ring. Or the spectrum is given functorially $X : cdga \rightarrow \infty$ -*

groupoid.

(definition 2 : homotopy)

The functorial perspective is that $X : sComm \rightarrow \infty - \text{groupoid}$ ($sComm$ or more generally E_∞ -algebras) Simplicial structure is preserved under ∞ -ness

Definition 0.0.6 (Stable ∞ -Category) Homotopy category of stable ∞ -category is triangulated category.

Definition 0.0.7 (Nerve) Nerve N is a simplicial set as $N(\mathcal{C})_0 = \{\text{objects of } \mathcal{C}\}$, and $N(\mathcal{C})_1 = \{\text{morphism of } \mathcal{C}\}$. For higher degree, $N(\mathcal{C})_2 = \{x \rightarrow y \rightarrow z \text{ for all objects } x, y, z \text{ and existing objects}\}$ is a composition of morphisms. No wonder $N(\mathcal{C})_3 = \{x \rightarrow y \rightarrow z \rightarrow w\}$.

Definition 0.0.8 (Classifying Space) For a group G , the classifying space BG is a space s.t. $[X, BG] \cong \{\text{principal } G\text{-bundles over } X\} / \sim$

(Functor)

$Fun(BG, \mathcal{C})$ is a group action of the category $\mathcal{C}$, and $F(*) = \mathcal{C}$, and every $g \in G$ gives a functor $F(g) = \mathcal{C} \rightarrow \mathcal{C}$.

(example)

$$\pi_1(BG) = G$$

(category)

Interpreting BG as a category, BG is a category with one object $*$ and $Hom_{BG}(*, *) = G$.

Definition 0.0.9 (fibration/cofibration)

1 Scheme

Definition 1.0.1 (Functor of Points) In affine scheme, the whole geometry can be described functorially. The spectrum is a functor $Spec : CRing^{op} \rightarrow AffSch$ for a category of affine schemes $AffSch \cong CRing^{op}$. Naturally, there is natural transformation $Hom_{AffSch^{op}}(Spec(A), Spec(R)) \cong Hom_{CRing}(A, R)$.

Scheme is not in the same way, however, but if scheme is locally isomorphic to affine scheme, then it can be described locally using a functor of points: that is for a scheme X , a functor of points is a functor $h_X : AffSch^{op} \rightarrow Set$ s.t. $h_X(Spec(A)) = Hom_{Sch}(Spec(A), X)$. This notion can naturally lift to define sheaf by gluing each functor of point. A sheaf $\mathcal{F}$ is $\mathcal{F}|_U = h_X|_U$ for any subscheme $U \subset X$ isomorphic to affine scheme. Categorically, this is an embedding $AffSch \rightarrow Sh(AffSch)$.

Deformation functor is functor of points $Def_X : Art \rightarrow Sets$ (even restricting from commutative ring $CRing \rightarrow Sets$ or $AffSch^{op} \rightarrow Sets$ to artinian ring, or even local artinian ring).

For a flat morphism $X \rightarrow S$, deformation is to find commutative diagram $X \rightarrow \mathfrak{X}$

$S \rightarrow B$ where this $\mathfrak{X}$ is not unique.

deformation functor only controls local deformation. by fomrality $Def_X(k) \cong Hom_{loc}(R, A)$

Definition 1.0.2 (Deformation Functor)

2 Hochschild

Definition 2.0.1 (Hochschild Chain Complex) (Algebra)

Let A be an associative algebra over a commutative ring k , and M is A -bimodule (equivalently left $A^e = A \otimes_k A^{op}$ -module). Then Hochschild cochain complex is $C_n(M, A) := M \otimes_k A^{\otimes n}$ with differential map $b : C_n(A, M) \rightarrow C_{n-1}(M, A)$ s.t. $b^2 = 0$. The cohomology class is given by $HH_n(A, M) = H_n(C(A, M))$. Equivalently, cohomology class is $HH^n(A, M) = Ext_{A^e}^n(A, M)$, while homology class is $HH_n(A, M) = Tor_n^{A^e}(A, M)$.

Definition 2.0.2 (Homotopy Category) Conventional meaning of homotopy for a topological space X is a classification of continuous functions $f : S^n \rightarrow X$, or $f \in Maps(S^n, X)$ by another notation, and $f, g \in Maps(S^n, X)$ is homotopy if there is a continuous transformation $S^n \times I \rightarrow X$. This means f, g are elements of the functional space $Maps(S^n, X)$, and they have a continuous path starting from f to g , which means connected. Hence $[f] = [g] \in \pi_0 Maps(X, Y)$.

Based on that data, we define homotopy category consists of $Hom_{hC}(X, Y) = \pi_0 Maps_C(X, Y)$ where $Maps(X, Y)$ is a mapping space, which means a set of continuous maps from X to Y , where objects X and Y are any topological spaces. Further, $\pi_n Maps(X, Y) = \pi_0(Maps(S^n, Maps(X, Y)))_f$.

Definition 2.0.3 (Stable ∞ Category) Let $\mathcal{C}$ be a stable ∞ category. Its homotopy category $h\mathcal{C}$ defined by $Hom_{h\mathcal{C}}(X, Y) = \pi_0 Maps_{\mathcal{C}}(X, Y)$. It's defined in the same way as the ordinary category, thus higher n -morphism is ignored for making this homotopy category. If in particular if $\mathcal{C}$ is derived ∞ category $D_{\infty}(A)$ for some abelian category A (derived ∞ category $D_{\infty}(A)$ is stable ∞ category not the other way), its homotopy category $hD_{\infty}(A)$ is derived category. Then $\pi_n Map_{D_{\infty}(A)}(X, Y) \cong Hom_{D(A)}(X, Y[n]) \cong Ext_A^n(X, Y)$ for $n \geq 0$.

Definition 2.0.4 (Hochschild Chain Complex Categorical) Let's consider categorification of algebraic case of Hochschild cohomology $HH^n(A, M) = Ext_{A^e}^n(A, M)$. $HH(\mathcal{C}) = REnd_{Fun(\mathcal{C}, \mathcal{C})}(Id_{\mathcal{C}}, Id_{\mathcal{C}})$

$$End(id_{Perf(A)}) \cong RHom_{A^e}(A, A)$$

Definition 2.0.5 (Perfect Complex) *Perfect complex is a compact object in unbounded derived category $D(A)$ of A -modules. An object X is compact if the functor $\text{Hom}_{\mathcal{C}}(X, -)$ commutes with colimits by natural transformation $\text{colim}(\text{Hom}_{\mathcal{C}}(X, Y_i) \rightarrow \text{Hom}_{\mathcal{C}}(X, \text{colim}_i(Y_i)))$ is bijection.*

In particular if A is noetherian, a module over A is perfect iff finitely generated and of finite projective dimension.

Definition 2.0.6 (Colored Operad) *An ordinary operad is a family $\{O(n)\}_{n \in \mathbb{N}}$ s.t. $O(n) = \{X^n \rightarrow X\}$ for some space X consists of n -ary operations, and it has a composition $O(n) \times O(k_1) \times \cdots \times O(k_n) \rightarrow O(\Sigma k_i)$. If $O(n)$ is a topological space for each n , then $\{O(n)\}$ is called topological operad.*

(Categorification)

Let $\mathcal{C}$ be an abelian symmetric monoidal k -linear category and closed under infinite sums and products. A polynomial functor $F : \mathcal{C} \rightarrow \mathcal{C}$ is defined by $F(V) := \bigoplus_{n \geq 0} (F(n) \otimes V^{\otimes n})_{S_n}$ where $F(n)$ is representation of symmetric group S_n . This polynomial functors F forms a category $\mathcal{PF}$ where morphisms are intertwiners $\text{Hom}(F, G) = \prod_{n \geq 0} \text{Hom}_{S_n}(F(n), G(n))$, and composition has natural isomorphism $(F \circ G)(V) \cong F(G(V))$. In addition, a unit functor 1 s.t. $1_1 = 1_{\mathcal{C}}$ and $1(n) = 0$ for all $n \neq 1$. An operad in $\mathcal{C}$ is a monoid in monoidal category $R \in \mathcal{PF}$ with morphism $m : R \circ R \rightarrow R$ and $u : 1 \rightarrow R$.

(E_n -operad)

little disk operad $\{D(k)\}$ consists of $D(k) = \{(D_1, \dots, D_k) | D_i \cap D_j = \emptyset, D_i \subset D^n\}$, which is a topological space, where D^n is a unit disk of dimension n . The dimension $D(k)$ could be k not necessarily the same as the dimension of the unit disk. This little disk operad makes E_n -algebra. Denoting $E_n(k) \times X^k \rightarrow X$ means the restriction of possible morphisms $X^k \rightarrow X$ by associativity of symmetry or any others.

(E_2 -operad)

$E_2(n) \times C.(A, A)^{\otimes n} \rightarrow C.(A, A)$ All Hochschild can be controlled by E_2 according to Deligne conjecture.

(Swiss Cheese operad)

SC operad is a colored operad defined over $I = \{c, o\}$, where $SC(c^k, o^l; o)$ is a notation that c (closed) has k inputs, o (open) has l inputs, and the output is o (open). In addition, $SC(c^k, o^l; o)$ is a topological operad, which means k -disks exist in upper-half disk of 2-dimension, and l -disks exist on the boundary. In particular $SC(c^k; c)(n)$ is $E_2(n)$ -operad. In particular $SC(o^k; o)(n)$ is $E_1(n)$ -operad.

(Colored operad)

For a set I , $\mathcal{C}^I$ is a category whose objects are (V_i) where $V_i \in \mathcal{C}$. A polynomial functor $F : \mathcal{C}^I \rightarrow \mathcal{C}^I$ is defined by $F((V_i)_{i \in I})_j = \bigoplus_{a: I \rightarrow \mathbb{Z}_{\geq 0}} F_{a, j} \otimes \prod S_{a(i)} \otimes_{i \in I} (V_i^{\otimes a(i)})$.

(Kontsevich-Soibelman operad)

It's also called open-closed operad, and its an algebraic operad, whose geometric analogy is Swiss-Cheese operad. Consider a colored operad that acts on $(C(A, A), A)$ where A is an A_∞ algebra. $C(A, A)^{\otimes m} \otimes A^{\otimes n} \rightarrow A$ mapping $(f, g, a, b, c) \mapsto f(g(a), b) + f(a, g(b))$ for $f \in C^2$ and $g \in C^1$ for $C^2 \otimes C^1 \otimes A^{\otimes 3} \rightarrow A$ (note every variable doesn't needs to be used so "c" is).

To generalize to KS operad, an A_∞ algebra $A^{\otimes n}$ to $C(A, A) = A \otimes A^{\otimes n}$, so the original data will be paraphrased to $(C(A, A), C(A, A))$ with morphisms $C(A, A)^{\otimes r} \otimes C(A, A) \rightarrow C(A, A)$ (note not $C(A, A)^{\otimes n}$ tensor product, but idk why). Here we define KS operad as an operad that acts on $(C(A, A), C(A, A))$.

3 Moduli Space

Fine moduli space is a scheme M but with a functor $F : Sch^{op} \rightarrow Set$ that is represented by M .

Formal moduli space $F : Art \rightarrow Set$ is moduli space but specific one.

Moduli stack is a stack $\mathfrak{X}$ but also $F : Schop \rightarrow Grpd$ or $F : Schop \rightarrow \infty - Grpd$, and it's represented by $\mathfrak{X}$, so $F(S) := Hom(S, \mathfrak{X})$.

References

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